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Editor-in-Chief
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Dear Editor,

I am pleased to submit my research manuscript for publication in *Microfluidics and Nanofluidics*, entitled “**A deformability threshold for stiffness-based sorting in deterministic-lateral-displacement microfluidic devices: a 2D lattice-Boltzmann immersed-boundary parameter sweep**”.

Deterministic lateral displacement (DLD) is the workhorse passive sorting technology of modern microfluidic cell-separation chips, but the founding Huang–Holm critical-bin theory was derived for rigid spheres. Experimental and computational studies have since shown that deformable particles can be sorted by stiffness at *identical particle size*, opening DLD to label-free mechanophenotyping of red blood cells, circulating tumour cells and synthetic vesicle drug carriers. What was missing from the literature is a quantitative answer to the design question every DLD chip designer faces: *at what Capillary number does stiffness-based sorting actually switch on?* Below that threshold the chip behaves geometrically (rigid-sphere DLD) and stiffness contrast does nothing for the user; above it, sorting fidelity grows monotonically with contrast. This manuscript identifies that threshold quantitatively.

The main contributions of this work are:

- **Custom lattice-Boltzmann immersed-boundary code with full discretisation documented:** all simulations were performed with SOFTFLOW, a C++ LBM-IBM code I developed in-house. The manuscript reports the complete numerical scheme — D2Q9 distribution functions, regularised BGK collision (Latt–Chopard), Guo body forcing, interpolated bounce-back (Bouzidi–Firdaouss–Lallemand) for curved pillars, multi-direct-forcing IBM (Luo et al.), and the Skalak constitutive law for the membrane — so that the results are reproducible from the equations alone.
- **Matched-membrane no-contrast diagonal control:** the two species in the sweep differ *only* in their surface shear modulus G_s ; bending, area and perimeter moduli are held identical. The diagonal of the sweep grid ($G_s^{\text{soft}} = G_s^{\text{stiff}}$) is therefore a strict no-contrast control in which the displacement-angle separation must vanish in expectation. To my knowledge no prior DLD-deformability study has imposed this kind of within-grid control.
- **Diagonal-subtracted angular separation $\Delta\theta_{\text{corr}}$:** I introduce a per-row baseline subtraction that removes the finite- N noise floor from the off-diagonal cells, isolating the pure deformability-contrast signal. The diagonal cells deliver an empirically determined noise band of $\sim 1.3^\circ$ that is then quantitatively subtracted from every off-diagonal measurement.

- **Identification of a Capillary-number threshold:** the sweep reveals that stiffness-based sorting emerges only when the soft-species Capillary number exceeds $Ca_{\text{soft}} \approx 0.2$. Below this threshold, the off-diagonal signal collapses to within the noise band regardless of stiffness ratio; above it, $|\Delta\theta_{\text{corr}}|$ grows monotonically up to a Capillary ratio of ~ 13 before saturating.
- **Practical chip-design rule:** the threshold $Ca_{\text{soft}} > 0.2$ translates directly into a minimum flow speed for any DLD chip targeting a population of given stiffness. For physiological red blood cells in a $5\mu\text{m}$ gap this corresponds to a centre-line velocity of order 1 mm/s; for stiffer circulating tumour cells the required speed scales linearly with stiffness and may begin to conflict with shear-induced cell-damage limits — a fundamental trade-off that this manuscript quantifies.
- **Provenance-rich open data:** every cell of the sweep ships a self-contained `run_manifest.json` (git commit, compiler identity, resolved CXX flags, OpenMP thread count, full parameters, RNG seed) that is sufficient to reproduce the cell bit-for-bit when combined with the equations in the Methods section.

The manuscript directly addresses the scope of *Microfluidics and Nanofluidics*: it combines novel numerical methodology, systematic parameter-space exploration, and an immediately useful design rule for microfluidic cell-separation chips. The approach is reproducible by any group with access to an LBM-IBM solver of comparable fidelity, and the methodological contributions (matched-membrane control, diagonal subtraction) are transferable to any deformability-vs-geometry trade-off study in related microfluidic separation techniques (inertial focusing, viscoelastic migration, pinched-flow fractionation).

This is a single-author manuscript that I have prepared in full. The work has not been published previously and is not under consideration for publication elsewhere. I have no conflicts of interest to declare. The work was conducted at the Department of Natural Sciences, The Begum Nusrat Bhutto Women University, Sukkur, with institutional support from the Sindh Higher Education Commission.

Thank you for considering this manuscript. I look forward to your response.

Sincerely,

Dr. Ram Chand